

Please check the examination details below before entering your candidate information

Candidate surname

mel@justmaths.co.uk

Other names

**Pearson Edexcel
International GCSE**

Centre Number

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Candidate Number

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Wednesday 15 January 2020

Morning (Time: 2 hours)

Paper Reference **4MA1/2H**

Mathematics A

Paper 2H

Higher Tier

WORKED
SOLUTIONS



You must have:

Ruler graduated in centimetres and millimetres, protractor, compasses, pen, HB pencil, eraser, calculator. Tracing paper may be used.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Without sufficient working, correct answers may be awarded no marks.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- **Calculators may be used.**
- You must **NOT** write anything on the formulae page.
Anything you write on the formulae page will gain NO credit.

Information

- The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

Turn over ►

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Pearson

International GCSE Mathematics

Formulae sheet – Higher Tier

Arithmetic series

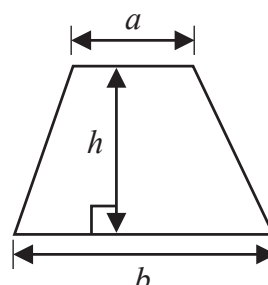
Sum to n terms, $S_n = \frac{n}{2} [2a + (n-1)d]$

The quadratic equation

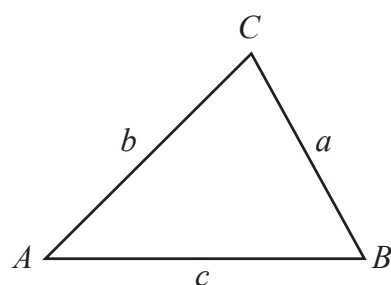
The solutions of $ax^2 + bx + c = 0$ where $a \neq 0$ are given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Area of trapezium = $\frac{1}{2}(a+b)h$



Trigonometry



In any triangle ABC

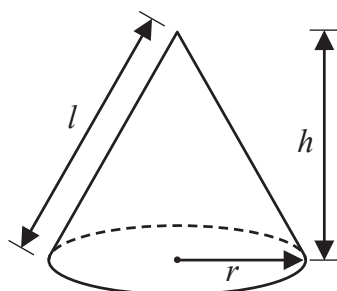
Sine Rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Cosine Rule $a^2 = b^2 + c^2 - 2bc \cos A$

Area of triangle = $\frac{1}{2}ab \sin C$

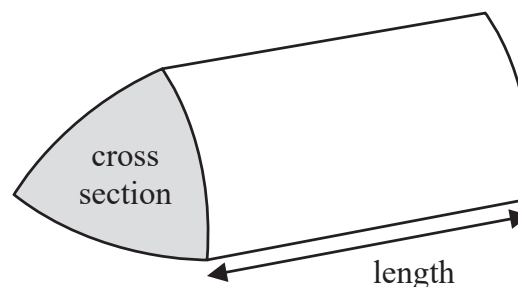
Volume of cone = $\frac{1}{3}\pi r^2 h$

Curved surface area of cone = $\pi r l$



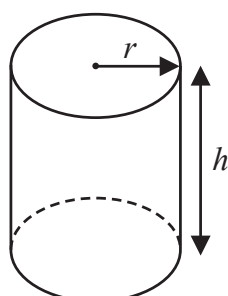
Volume of prism

= area of cross section $\times$ length



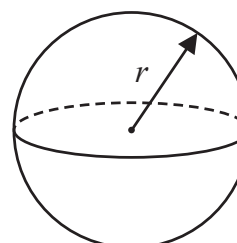
Volume of cylinder = $\pi r^2 h$

Curved surface area of cylinder = $2\pi r h$



Volume of sphere = $\frac{4}{3}\pi r^3$

Surface area of sphere = $4\pi r^2$



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Answer ALL TWENTY SIX questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 (a) Simplify $\frac{x^9}{x^2}$

$$x^{9-2}$$

$$\frac{x^7}{\dots\dots\dots} \quad (1)$$

(b) Write $\frac{7^8 \times 7^4}{7^3}$ as a single power of 7

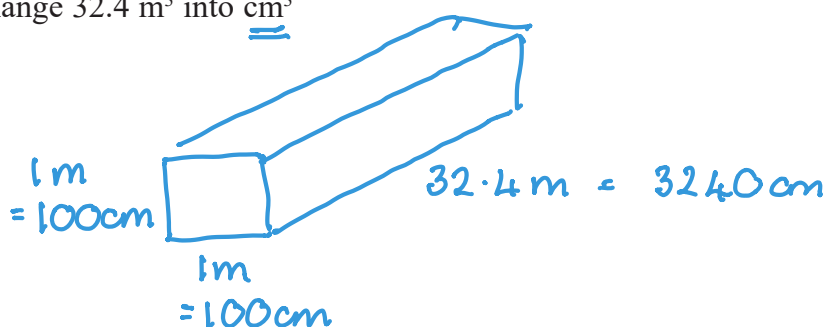
$$7^{8+4} = 7^{12}$$

$$7^{12-3} = 7^9$$

$$\frac{7^9}{\dots\dots\dots} \quad (2)$$

(Total for Question 1 is 3 marks)

2 Change 32.4 m^3 into cm^3



$$3240 \times 100 \times 100$$

$$\frac{32400000}{\dots\dots\dots} \text{ cm}^3$$

(Total for Question 2 is 2 marks)



3 Show that $4\frac{2}{3} + 3\frac{4}{5} = 8\frac{7}{15}$

$$4\frac{2}{3} = \frac{14}{3}$$

$$3\frac{4}{5} = \frac{19}{5}$$

$$\begin{array}{ccc} \frac{14}{3} & + & \frac{19}{5} \\ \downarrow \times 5 & & \downarrow \times 3 \\ \frac{70}{15} & + & \frac{57}{15} & = & \frac{127}{15} \end{array}$$

$$\frac{127}{15} = 8\frac{7}{15}$$

$$\begin{aligned} 15 \times 8 &= 120 \\ 127 - 120 &= 7 \end{aligned}$$

(Total for Question 3 is 3 marks)

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- 4 The diagram shows a triangle.

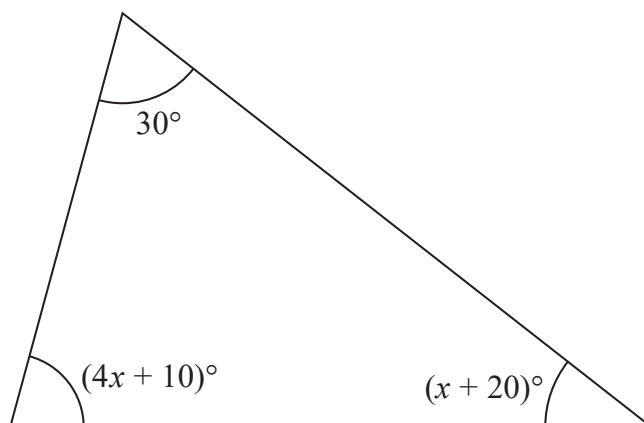


Diagram **NOT**
accurately drawn

Work out the value of x .

$$180 = 30 + 4x + 10 + x + 20$$

$$180 - 60 = 5x$$

$$x = \frac{120}{5}$$

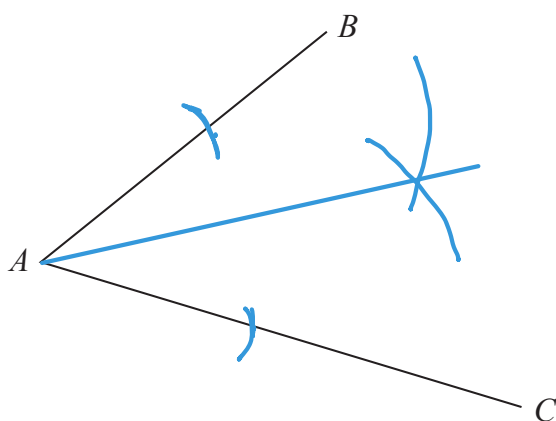
$$= 24$$

$$x = \underline{\quad 24 \quad}$$

(Total for Question 4 is 4 marks)



- 5 Use ruler and compasses to construct the bisector of angle BAC .
You must show all your construction lines.



(Total for Question 5 is 2 marks)



- 6 A bag contains only red beads, blue beads, green beads and yellow beads.

The table gives the probabilities that, when a bead is taken at random from the bag, the bead will be blue or the bead will be yellow.

Colour	red	blue	green	yellow
Probability	0.15	0.24	0.3	0.31

The probability that the bead will be green is twice the probability that the bead will be red.

Sofia takes at random a bead from the bag.

She writes down the colour of the bead and puts the bead back into the bag.

She does this 180 times.

Work out an estimate for the number of times she takes a red bead from the bag.

$$1 - (0.24 + 0.31) \\ = 1 - 0.55 = 0.45$$

$$0.45 \div 3 = 0.15$$

$$\begin{array}{cc} R & G \\ 0.15 & 0.3 \\ \times 2 & \nearrow \end{array}$$

Estimate $0.15 \times 180 = 27$

27

(Total for Question 6 is 4 marks)



- 7 (a) Solve the inequality $2x + 7 > 4$

$$-7 \quad -7$$

$$2x > -3$$

$$x > -\frac{3}{2}$$

$$x > -\frac{3}{2}$$

(2)

- (b) Solve $x^2 - 3x - 40 = 0$
Show clear algebraic working.

$$5, 8$$

$$5 - 8$$

$$(x + 5)(x - 8) = 0$$

$$x + 5 = 0$$

$$x = -5$$

$$x - 8 = 0$$

$$x = 8$$

$$x = -5 \quad x = 8$$

(3)

(Total for Question 7 is 5 marks)



- 8 The table shows the cost, in euros, of Brigitte's car insurance in each of the years 2016, 2017 and 2018

Year	2016	2017	2018
Cost of insurance (euros)	500	545	592

Brigitte says,

"The percentage increase in the cost of my car insurance from 2017 to 2018 is more than the percentage increase in the cost of my car insurance from 2016 to 2017"

- (a) Is Brigitte correct?

You must show how you get your answer.

2016 → 2017

increase = 45

$$\begin{aligned} \% &= \frac{45}{500} \times 100 \\ &= 9 \end{aligned}$$

2017 → 2018

increase = 47

$$\begin{aligned} \% &= \frac{47}{545} \times 100 \\ &= 8.623... \end{aligned}$$

$$8.6 < 9$$

Brigitte is not correct.

(4)

Henri wants to insure his car.

He gets a discount of 15% off the normal price.

Henri pays 952 euros for his car insurance after the discount.

- (b) Work out the discount that Henri gets.

$$\begin{array}{lcl} 85\% & = & 952 \\ \div 85 \downarrow & & \downarrow \div 85 \\ 1\% & = & 11.2 \\ \times 100 \downarrow & & \downarrow \times 100 \\ 100\% & = & 1120 \end{array}$$

$$1120 - 952 = 168$$

168

euros

(3)

(Total for Question 8 is 7 marks)



- 9 The density of gold is 19.3 g/cm^3
A gold bar has volume 150 cm^3

Work out the mass of the gold bar.

$$D = \frac{m}{V}$$

$$m = D \times V$$

$$\begin{aligned} \text{mass} &= 19.3 \times 150 \\ &= 2895 \end{aligned}$$

2895

g

(Total for Question 9 is 2 marks)

- 10 Change a speed of 50 metres per second to a speed in kilometres per hour.

$$\begin{aligned} 50 \text{ m} &= 1 \text{ s} \\ \downarrow &\times 3600 \\ 180000 \text{ m} &= 3600 \text{ s} \end{aligned}$$

$$\begin{aligned} 180000 \text{ metres in 1 hour} \\ \div 1000 \end{aligned}$$

$$= 180 \text{ km} = 1 \text{ hr}$$

so 180 km per hour

180

kilometres per hour

(Total for Question 10 is 3 marks)

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- 11 The diagram shows a shaded shape $ABCD$ made from a semicircle ABC and a right-angled triangle ACD .

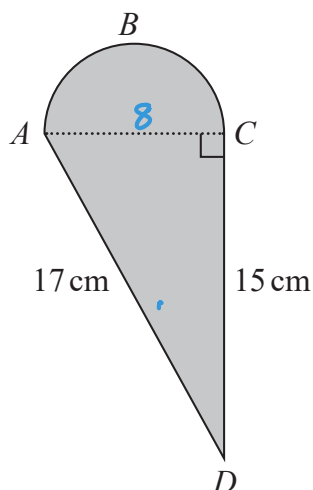


Diagram **NOT**
accurately drawn

AC is the diameter of the semicircle ABC .

Work out the perimeter of the shaded shape.

Give your answer correct to 3 significant figures.

$$AC^2 = 17^2 - 15^2$$

$$= 64$$

$$AC = \sqrt{64} = 8$$

Semicircle ABC $d = 8$

$$\frac{1}{2} \pi \times 8 = 4\pi$$

$$= 12.566\dots$$

$$\text{Perimeter} = 17 + 15 + 12.566\dots$$

$$= 44.566\dots$$

↑
(3 sf.)

44.6

..... cm

(Total for Question 11 is 5 marks)



- 12 Astrid wants to buy some oil.
She can buy the oil from either Dane Oil or Arctic Oil.

Here is information about the price that each company will charge Astrid.

Dane Oil	Arctic Oil
(4.2×10^5) litres for 2 500 000 Krone	(8.6×10^5) litres for 770 000 Dollars

$$770000 \times 6.57 = 5058900 \text{ krone}$$

Astrid wants to get the better value for money for the oil.

$$1 \text{ Dollar} = 6.57 \text{ Krone}$$

From which company should she buy her oil, Dane Oil or Arctic Oil?
You must show your working.

$$\begin{array}{ll} \text{Dane .} & \text{Arctic Oil} \\ \text{Krone} \div \text{litres} & \\ = 5.95... & = 5.88... \end{array}$$

Arctic Oil is better value

(Total for Question 12 is 4 marks)



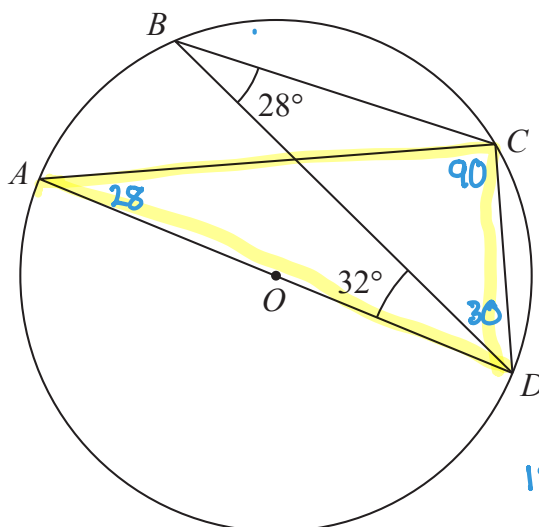


Diagram **NOT**
accurately drawn

$$180 - (90 + 60) = 180 - 150$$

A, B, C and D are points on a circle, centre O .
 AOD is a diameter of the circle.

Angle $CBD = 28^\circ$

Angle $BDA = 32^\circ$

Find the size of angle BDC .

Give a reason for each stage of your working.

$$ACD = 90 \quad \text{angle in a semicircle is } 90$$

$$BAC = 28^\circ \quad \text{angles in the same segment are equal}$$

$$\begin{aligned} ADC &= 180 - (28 + 90 + 32) \quad \text{angles in a triangle} = 180^\circ \\ &= 180 - 150 \\ &= 30 \end{aligned}$$

30

(Total for Question 13 is 4 marks)

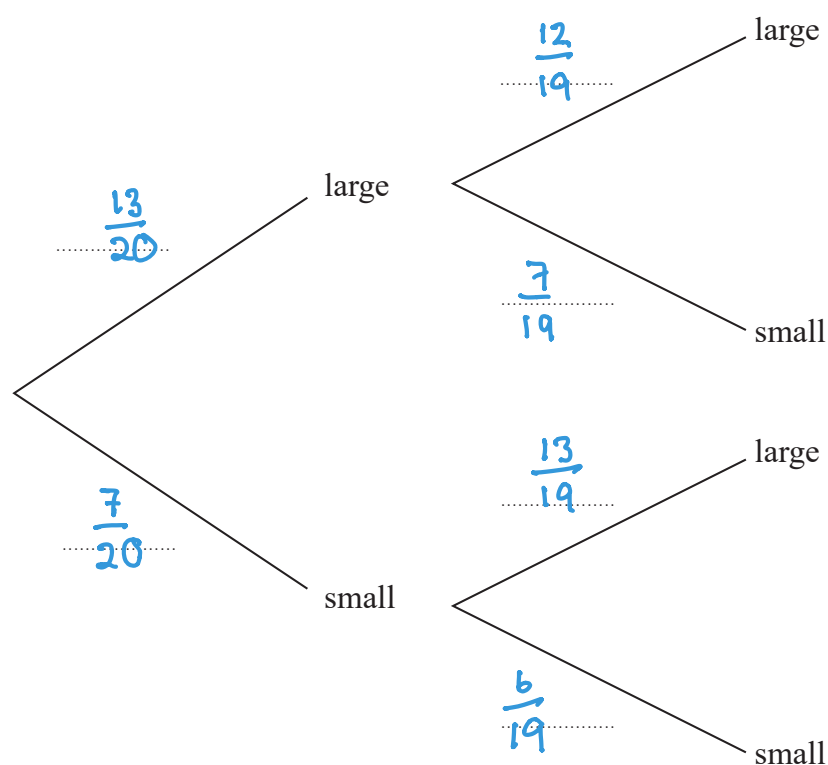


14 There are 20 glasses in a cupboard.

13 of the glasses are large
7 of the glasses are small

Roberto takes at random two glasses from the cupboard.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Roberto takes two small glasses.

$$P(S,S) = \frac{7}{20} \times \frac{6}{19}$$

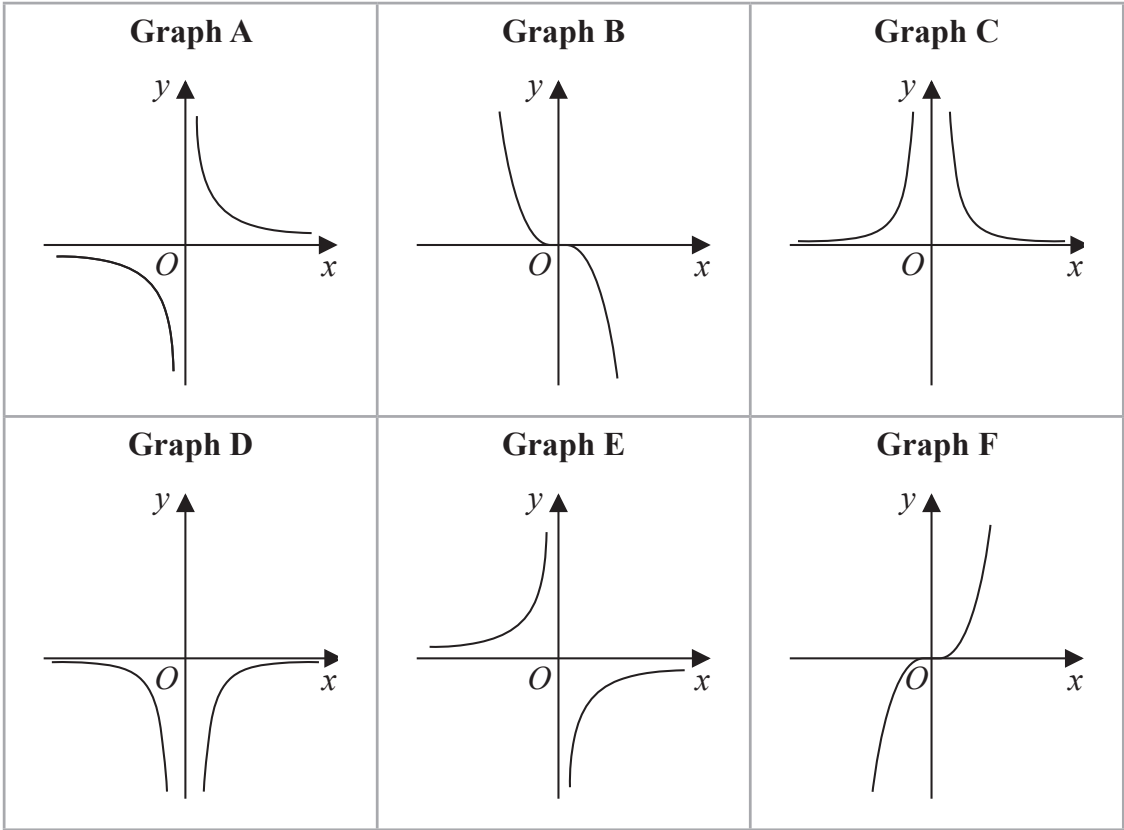
$$\frac{42}{380} = \frac{21}{190}$$

(2)

(Total for Question 14 is 4 marks)



15 Here are six graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \frac{2}{x^2}$	<u> C </u>
$y = -\frac{1}{2}x^3$	<u> B </u>
$y = -\frac{5}{x}$	<u> E </u>

(Total for Question 15 is 3 marks)

16 Make x the subject of $y = \sqrt{\frac{x+1}{x-4}}$

$$y^2 = \frac{x+1}{x-4}$$

$$y^2(x-4) = x+1$$

$$y^2x - 4y^2 = x+1$$

$$y^2x - x = 4y^2 + 1$$

$$x(y^2 - 1) = 4y^2 + 1$$

$$x = \frac{4y^2 + 1}{y^2 - 1}$$

$$x = \frac{4y^2 + 1}{y^2 - 1}$$

(Total for Question 16 is 4 marks)

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- 17 Prove that the difference between two consecutive square numbers is always an odd number.
Show clear algebraic working.

n = any number

$n+1$ = next number

$$\begin{aligned}(n+1)^2 - n^2 &= n^2 + 2n + 1 - n^2 \\ &= 2n + 1\end{aligned}$$

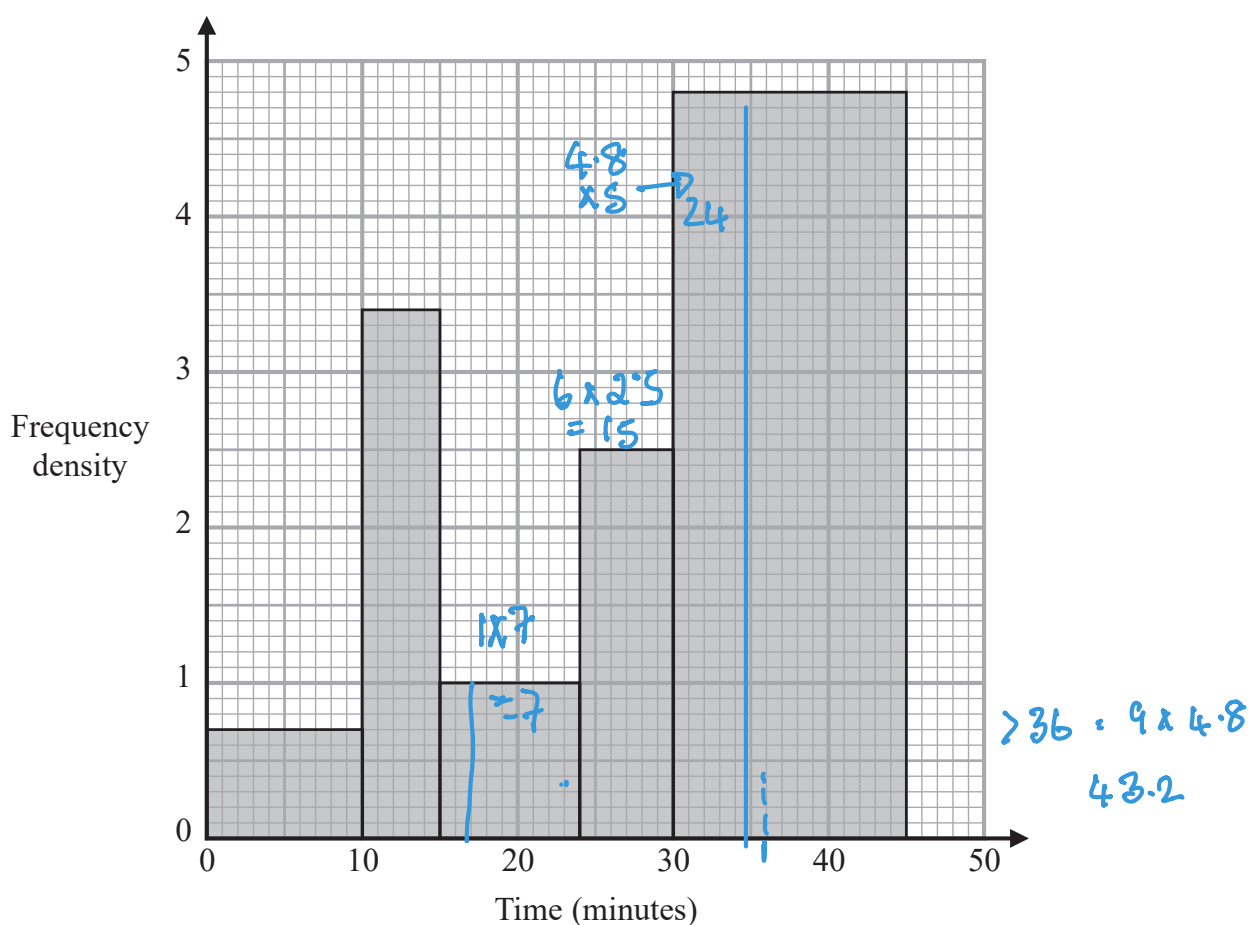
$2n$ is always even

$\therefore 2n+1$ will always be odd

(Total for Question 17 is 3 marks)



- 18 The histogram gives information about the times, in minutes, that some customers spent in a supermarket.



- (a) Work out an estimate for the proportion of these customers who spent between 17 minutes and 35 minutes in the supermarket.

$$\begin{aligned}
 \text{Total} &= 0.7 \times 10 + 3.4 \times 5 + 9 \times 1 + 2.5 \times 6 + 4.8 \times 5 \\
 &= 7 + 17 + 9 + 15 + 24 \\
 &= 72
 \end{aligned}$$

$$\begin{aligned}
 \underline{17 \rightarrow 35} \quad & 1 \times 7 + 6 \times 2.5 + 4.8 \times 5 \\
 & 7 + 15 + 24 = 46
 \end{aligned}$$

$$\begin{array}{r}
 46 \\
 \hline
 120
 \end{array}$$

(3)



One of the customers is selected at random.

Given that this customer had spent more than 30 minutes in the supermarket, **72**

(b) find the probability that this customer spent more than 36 minutes in the supermarket.

$$4.8 \times 9 = 43.2$$

$$\frac{43.2}{72} = \frac{3}{8} = \frac{9}{24}$$

$$\frac{3}{5} \text{ or equivalent}$$

(2)

(Total for Question 18 is 5 marks)

19 (a) Write down an equation of a line that is parallel to the line with equation $y = 7 - 4x$

$$\text{eg. } y = 6 - 4x$$

(1)

The line **L** passes through the points with coordinates $(-3, 1)$ and $(2, -2)$

(b) Find an equation of the line that is perpendicular to **L** and passes through the point with coordinates $(-6, 4)$

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

gradient of L

$$\frac{-2-1}{2-(-3)} = -\frac{3}{5}$$

perp line = $\frac{5}{3}$

$$y = \frac{5}{3}x + c \quad (-6, 4) \text{ on line}$$

$$4 = \frac{5}{3} \times -6 + c$$

$$4 + 10 = c \quad c = 14$$

$$\text{so } y = \frac{5}{3}x + 14$$

$$5x - 3y + 42 = 0$$

(4)

(Total for Question 19 is 5 marks)



20 The area of a rectangle is 18 cm^2

The length of the rectangle is $(\sqrt{7} + 1) \text{ cm}$.

Without using a calculator and showing each stage of your working,

find the width of the rectangle.

Give your answer in the form $a\sqrt{b} + c$ where a , b and c are integers.

$$18 = \sqrt{7} + 1 \times w$$

$$w = \frac{18}{1 + \sqrt{7}} \times \frac{1 - \sqrt{7}}{1 - \sqrt{7}} = \frac{18 - 18\sqrt{7}}{1 - 7}$$

$$= \frac{18 - 18\sqrt{7}}{-6}$$

$$= -\frac{18}{6} + \frac{18\sqrt{7}}{6}$$

$$= -3 + 3\sqrt{7}$$

$$= 3\sqrt{7} - 3$$

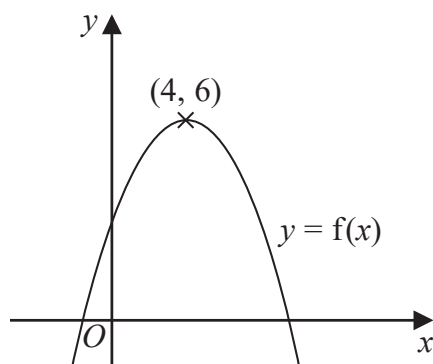
$$3\sqrt{7} - 3$$

..... cm

(Total for Question 20 is 3 marks)



21 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.

The coordinates of this maximum point are (4, 6)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 4)$ ←

(.....0....., 6.....)

(ii) $y = f(2x)$

(.....2....., 6.....)
(2)

The equation of a curve **C** is $y = x^2 + 3x + 4$

The curve **C** is transformed to curve **S** under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$ (x-4)

(b) Find an equation of curve **S**.

You do not need to simplify the equation.

$$\begin{aligned} y - 6 &= (x - 4)^2 + 3(x - 4) + 4 \\ &= x^2 - 8x + 16 + 3x - 12 + 4 + 6 \\ y &= x^2 - 5x + 14 \end{aligned}$$

$$y = x^2 - 5x + 14$$

(2)

(Total for Question 21 is 4 marks)



- 22 The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points A and B .

Find the coordinates of A and B .

Show clear algebraic working.

$$y = x + 2 \quad y^2 = x^2 + 4x + 4$$

$$x^2 + x^2 + 4x + 4 - 2(x + 2) - 24 = 0$$

$$2x^2 + 4x + 4 - 2x - 4 - 24 = 0$$

$$2x^2 + 2x - 24 = 0$$

$$x^2 + x - 12 = 0$$

$$(x + 4)(x - 3) = 0$$

$$x = -4$$

$$x = 3$$

$$y = -4 + 2 \\ = -2$$

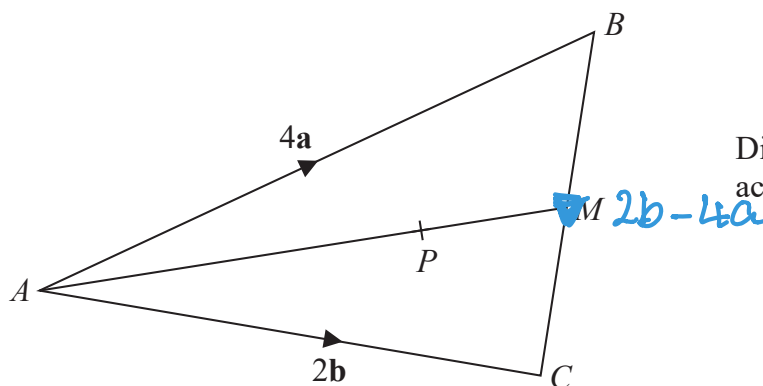
$$y = 3 + 2 \\ = 5$$

$$(-4, -2)$$

$$(3, 5)$$

(Total for Question 22 is 5 marks)





ABC is a triangle.

The midpoint of BC is M .

P is a point on AM .

$$\vec{AB} = 4\mathbf{a}$$

$$\vec{AC} = 2\mathbf{b}$$

$$\vec{AP} = \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$$

Find the ratio $AP:PM$

$$\vec{BC} = 2\mathbf{b} - 4\mathbf{a}$$

$$\vec{BM} = \mathbf{b} - 2\mathbf{a}$$

$$\vec{AM} = 4\mathbf{a} + \frac{1}{2}(2\mathbf{b} - 4\mathbf{a})$$

$$= 4\mathbf{a} - 2\mathbf{a} + \mathbf{b}$$

$$= 2\mathbf{a} + \mathbf{b}$$

$$\vec{PM} = -\frac{3}{2}\mathbf{a} - \frac{3}{4}\mathbf{b} + 4\mathbf{a} + \mathbf{b} - 2\mathbf{a} = \frac{1}{2}\mathbf{a} + \frac{1}{4}\mathbf{b}$$

$$AP : PM$$

$$1.5\mathbf{a} + 0.75\mathbf{b} : 0.5\mathbf{a} + 0.25\mathbf{b}$$

$$3 : 1$$

$$3 : 1$$

(Total for Question 23 is 3 marks)



$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

$$\begin{aligned} & \rightarrow x(9-4x^2) \\ & = x(3-2x)(3+2x) \\ & \leftarrow (3x-1)(2x-5) \end{aligned}$$

as a single fraction in its simplest form.

$$\frac{4(2x-3) - 3(2x-5)}{(2x-5)(2x-3)} \times \frac{(3x-1)(2x-5)}{x(3-2x)(3+2x)}$$

$$\Rightarrow \frac{8x-12-6x+15}{(2x-5)(2x-3)} \times \frac{(3x-1)\cancel{(2x-5)}}{x(3-2x)(3+2x)}$$

$$\Rightarrow \frac{\cancel{2x+3}}{(2x-3)} \times \frac{(3x-1)}{x(3-2x)\cancel{(3+2x)}}$$

$$\Rightarrow \frac{3x-1}{x(3-2x)(2x-3)}$$

$$\frac{3x-1}{x(3-2x)(2x-3)}$$

(Total for Question 24 is 4 marks)



25 Mario is going to save \$50 in the year 2021

He is going to continue to save, up to and including the year 2070, by increasing the amount he saves each year by \$ k

Mario will save a total of \$33 125 from 2021 to 2070

Work out the value of k .

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

$$33125 = \frac{50}{2} (2 \times 50 + (49) \times k)$$

$$\frac{33125}{25} = 100 + 49k$$

$$1325 = 100 + 49k$$

$$49k = 1225$$

$$k = \frac{1225}{49}$$

$$= 25$$

$$k = 25$$

(Total for Question 25 is 3 marks)



26 Here is a sector, AOB , of a circle with centre O and angle $AOB = x^\circ$

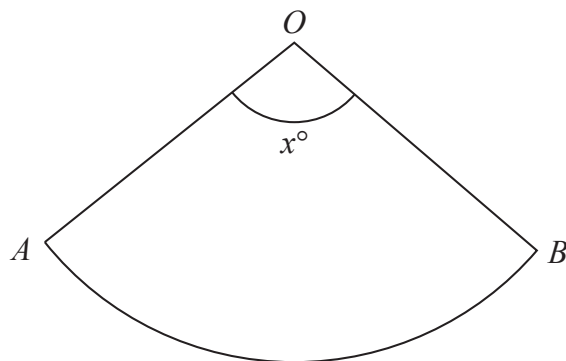


Diagram **NOT** accurately drawn

The sector can form the curved surface of a cone by joining OA to OB .

$$\frac{1}{3}\pi r^2 h$$

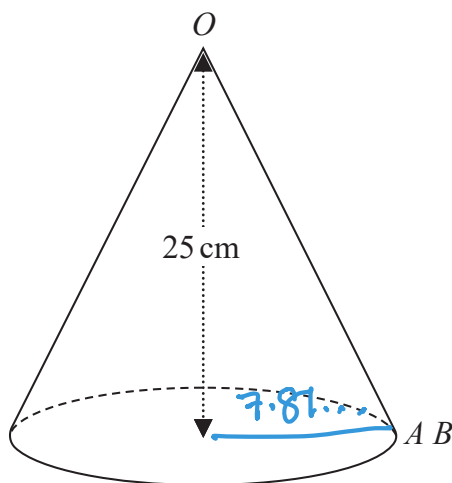


Diagram **NOT** accurately drawn

The height of the cone is 25 cm.

The volume of the cone is 1600 cm^3

Work out the value of x .

Give your answer correct to the nearest whole number.

CONE

$$1600 = \frac{1}{3} \times \pi \times r^2 \times 25$$

$$\sqrt{\frac{1600 \times 3}{25 \pi}} = r = 7.8176...$$

OB

$$\begin{aligned} OB^2 &= 7.8176...^2 + 25^2 \\ &= 686.115... \end{aligned}$$

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



$$OB = \sqrt{686.115...}$$

$$= 26.1938...$$

surface area of cone = $\pi r l$

$$= \pi \times 7.8176 \times 26.1938$$

$$= 643.31$$

sector.

$$\frac{x}{360} \times \pi \times 26.19...^2 = 643.3...$$

$$x = \frac{643.3... \times 360}{26.19...^2 \times \pi}$$

$$= 107.4...$$

$$x = 107$$

(Total for Question 26 is 6 marks)

TOTAL FOR PAPER IS 100 MARKS

